

Adaptive cruise control (ACC).

ACC allows users to relax while driving, especially on highways, by automatically adjusting the car's speed by matching the speed of the vehicles in front. The IMU utilised in the ACC system maintains a safe distance between vehicles by measuring the vehicle's acceleration and deceleration. If a change in speed is encountered in the lead vehicle, inertial sensors enable the system to respond quickly, thus promoting safe and comfortable driving.

Lane departure warning (LDW).

The LDW system provides users with a stress-free parking experience in crowded garages by minimising concerns about collision-related damage to the vehicle. While driving, the LDW system ensures safety by warning drivers about leaving the marked lanes without giving proper indicators. The inertial sensors monitor the vehicle's lateral motion and orientation and detect when the car unintentionally drifts out of its lane. Fast alerts are triggered, avoiding collisions.

Electronic stability control (ESC). ESC is a crash avoidance system which activates automatically when loss of control over the vehicle is detected. The system applies the brakes on individual wheels, preventing the car from skidding and avoiding severe accidents. Inertial sensors continuously monitor the vehicle's acceleration, angular velocity, and orientation. The system can then selectively apply braking force to individual wheels and adjust engine torque to help the driver maintain control during movements like sharp turns. IMU helps detect vehicle rollover by activating stability control measures or issuing drivers warnings, thereby ensuring safety.

Autonomous emergency braking (AEB). As the name suggests, AEB comes into action in critical situations. AEB system's primary function is to turn crashes into close calls. In

Understanding and utilising IMUs

An IMU is typically a combination of accelerometers, gyroscopes, and magnetometers. Magnetometers are optional.

- The accelerometers measure the specific force/acceleration of the body in different directions.
- The gyroscopes measure the angular rate of the body around different axes.
- When included, the magnetometers measure the magnetic field around the system and determine the body's orientation relative to the Earth's magnetic field.
- Combining the readings from these sensors, an IMU provides complete information about the motion and orientation of a body in real time.

With their ability to deliver accurate measurements, IMUs are well-suited for tasks demanding precision, such as medical applications, robotics, motion capture, and so on. The low power consumption of IMU makes them ideal for battery-powered applications. The cost-effectiveness positions them as an affordable solution across various domains.

short, it recognises the dangers and applies the brakes for the driver. Inertial sensors are utilised in AEB systems to detect potential collisions and assist in avoiding or mitigating their severity. Responsiveness and accuracy of ABE systems are enhanced using inertial sensors by providing data to the system and helping to make decisions.

Although not all ADAS come with all these features, several car manufacturers, including Tesla, BMW, Hyundai, Audi, Honda, Kia and many more, offer vehicles with ADAS systems that incorporate a comprehensive set of common features.

Advancement in IMU-based ADAS functionalities

In recent years, several advancements in ADAS, including IMUs, led to notable improvements in the performance, accuracy, and safety of ADAS functionalities. Some of the advances in this technology comprise the following.

Sensor fusion. Sensor fusion involves information from multiple sources like vehicle position, orientation, and sensor error characteristics and maintaining an updated record. IMUs provide valuable information on the vehicle's movement and orientation that can be combined with data from other sensors, such as cameras, lidar, and radar, to create a more comprehensive understanding of the vehicle's surroundings.

For example, IMU provides information on the vehicle's acceleration and angular velocity, yielding a more accurate environmental map when combined with lidar data. In sensor fusion, many sensors are involved. Still, IMU is the key sensor as it can blend the characteristics of all other sensors and, with simple math, come up with integrated results. The IMU, along with other sensors in autonomous vehicles, can better determine the position, speed, and trajectory of objects in the vehicle's vicinity and take appropriate actions to avoid collisions.

Triple redundant IMU in ADAS.

A triple redundant IMU is used in ADAS to enhance reliability and accuracy through redundancy. IMU triple redundancy utilises three separate IMUs to create a sensor architecture providing its measurements of acceleration, angular rate, and orientation. These IMUs operate independently and are cross-checked against each other to identify any inconsistencies or discrepancies in the data. By comparing the outputs from the redundant IMUs, the system can detect sensor faults or anomalies and take appropriate actions. Bosch utilises triple redundant IMUs in their ADAS systems to enhance the reliability and accuracy of motion sensing. Many automotive technology providers, like Aptiv, Aptiv, etc, also use triple redundant IMUs in